



Entergy Operations, Inc.
P.O. Box 756
Port Gibson, Mississippi 39150

Jeffery A. Hardy
Manager Regulatory Assurance
Grand Gulf Nuclear Station
Tel: 802-380-5124

10 CFR 50.73

GNRO2025-00025

August 18, 2025

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2025-003-00,
Manual Reactor Scram due to Degraded Main Condenser Vacuum
Caused by Circulating Water Pump Trip

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report (LER) 2025-003-00, Manual Reactor Scram due to Degraded Main Condenser Vacuum Caused by Circulating Water Pump Trip. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(iv)(A), for any event that resulted in manual actuation of the Reactor Protection System

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact me at 802-380-5124.

Sincerely,

A handwritten signature in blue ink, appearing to read 'JAH2'.

JAH/dr

Attachments: Licensee Event Report 2025-003-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

GNRO2025-00025

Page 3 of 3

Attachment
Licensee Event Report 2025-003-00

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027						
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)		Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.								
1. Facility Name Grand Gulf Nuclear Station, Unit 1		☒ 050 ☐ 052		2. Docket Number 00416		3. Page 1 of 2						
4. Title Manual Reactor Scram due to Degraded Main Condenser Vacuum Caused by Circulating Water Pump Trip												
5. Event Date			6. LER Number			7. Report Date		8. Other Facilities Involved				
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name N/A	☐ 050	Docket Number	
06	17	2025	2025	003	00	08	18	2025	Facility Name N/A	☐ 052	Docket Number	
9. Operating Mode 1						10. Power Level 70						
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)												
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)		
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)		
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)		
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)		
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)		
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)(i)		☐ 73.1200(f)		
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)		
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)		
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)						
☐ Other (Specify here, in abstract, or in NRC 366A).												
12. Licensee Contact for this LER												
Licensee Contact Jeff Hardy, Regulatory Assurance Manager								Phone Number (Include Area Code) 802-380-5124				
13. Complete One Line for each Component Failure Described in this Report												
Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS			
X	KE	TI		Y								
14. Supplemental Report Expected								15. Expected Submission Date		Month	Day	Year
☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)												
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)												
On June 17, 2025, Grand Gulf Nuclear Station (GGNS) Unit 1 was in MODE 1 at 70 percent power with all safety systems operable. At 04:26 CT, a manual scram was initiated due to degraded main condenser vacuum caused by the loss of the 'A' Circulating Water Pump. The event was caused by a failure of an embedded winding Resistance Temperature Device (RTD) in the Circulating Water Pump 'A' Motor which resulted in false high winding temperatures and actuation of the motor thermal overload protection function. All control rods fully inserted, and no radiological release occurred. Immediately after the scram, an expected reactor water Level 3 isolation signal was received. The reactor level was maintained by main feedwater. Actions were taken to replace the failed RTD and update the trip logic in order to prevent recurrence. This report is being made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for any event that resulted in manual actuation of the Reactor Protection System.												

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Grand Gulf Nuclear Station, Unit 1	☒ 050	00416	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2025	003	00

NARRATIVE**Plant Conditions:**

At the time of the event, Grand Gulf Nuclear Station (GGNS) Unit 1 was in MODE 1 at 70 percent power and all safety systems were operable.

Description of Event:

On June 17, 2025, at 04:26 CT, a manual scram was initiated due to degraded main condenser vacuum [SG] caused by the loss of the 'A' Circulating Water Pump [KE:P]. All control rods fully inserted and there were no complications. Immediately after the scram, an expected reactor water Level 3 isolation signal was received. The reactor level was maintained by main feedwater [SJ].

No other safety systems actuated, and no radiological release occurred.

Reportability:

This report is being made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for any event that resulted in manual actuation of the Reactor Protection System.

Cause(s):

The direct cause of the event was a failure of an embedded winding Resistance Temperature Device (RTD) in the Circulating Water Pump A Motor which resulted in false high winding temperatures and actuation of the motor thermal overload protection function.

Corrective Actions:

Completed Actions:

- The failure was corrected by removing the failed RTD from the circuit and utilizing an installed spare.
- The thermal overload trip logic was changed from a "1 out of 2" configuration to a "2 out of 2" configuration.

Safety Significance:

There were no actual safety consequences associated with this condition. Plant systems responded as designed.

Previous Similar Events:

LER 2025-001-00, Manual Scram Due to Degraded Condenser Vacuum, described a manual scram due to degrading condenser vacuum and loss of seal steam caused by a loss of power to the static inverter. The loss of power to the static inverter was due to rainwater intrusion to load control center.